

PRODUCT LITERATURE

EMECON TABLET 10MG

Each tablet contains:

Metoclopramide Hydrochloride 10 mg

EMECON SYRUP 5MG / 5 ML

Each 5ml contains:

Metoclopramide Hydrochloride 5 mg

Preservative in syrup

Methyl Paraben 0.11% w/v

Propyl Paraben 0.015% w/v

Description

Syrup: Yellow coloured clear syrup with sweet taste fruity odour.

Tablet: Round white odourless tablet with score on one side and plain on other side

Pharmacodynamics Properties

Metoclopramide hydrochloride is a substituted benzamide which stimulates the motility of the upper gastro-intestinal tract without affecting gastric acid secretion.

Metoclopramide hydrochloride increases gastric peristalsis leading to accelerated gastric emptying. Duodenal peristalsis is also increased which decreases intestinal transit time. The resting tone of the gastro-oesophageal sphincter is increased and the pyloric sphincter is relaxed.

Metoclopramide possesses parasympathomimetic activity as well as being a dopamine receptor antagonist with a direct effect on the Chemoreceptor trigger zone. Metoclopramide also increases prolactin secretion.

Pharmacokinetics Properties

Renal impairment

The clearance of metoclopramide is reduced by up to 70% in patients with severe renal impairment, while the plasma elimination half-life is increased (approximately 10 hours for a creatinine clearance of 10-50 mL/minute and 15 hours for a creatinine clearance <10 mL/minute).

Hepatic impairment

In patients with cirrhosis of the liver, accumulation of metoclopramide has been observed, associated with a 50% reduction in plasma clearance.

Indication

Adult population

Emecon is indicated in adults for:

- Prevention of delayed chemotherapy induced nausea and vomiting (CINV)
- Prevention of radiotherapy induced nausea and vomiting (RINV)
- Symptomatic treatment of nausea and vomiting, including acute migraine induced nausea and vomiting

Paediatric population

Emecon is indicated in children (aged 1 – 18 years) for:

- Prevention of delayed chemotherapy induced nausea and vomiting (CINV) as a second line option.

Route of administration

Oral.

Recommended Dose

Adults:

- The recommended single dose is 10mg, repeated up to three times daily.
- The maximum recommended daily dose is 30 mg or 0.5mg/kg body weight.
- The maximum recommended treatment duration is 5 days.

Prevention of delayed chemotherapy induced nausea and vomiting (CINV) (paediatric patients aged 1 – 18 years)

The recommended single dose is 0.1 to 0.15 mg/kg body weight, repeated up to three times daily by oral route. The maximum dose in 24 hours is 0.5 mg/kg body weight

Dosing Table:

Age	Body Weight	Dose	Frequency
1 – 3 years	10 – 14 kg	1 mg	Up to 3 times daily
3 – 5 years	15 – 19 kg	2 mg	Up to 3 times daily

5 – 9 years	20 -29 kg	2.5 mg	Up to 3 times daily
9 – 18 years	30 – 60 kg	5 mg	Up to 3 times daily
15 – 18 years	Over 60 kg	10 mg	Up to 3 times daily

The maximum treatment duration is 5 days for prevention of delayed chemotherapy induced nausea and vomiting (CINV).

Tablets are not suitable for use in children weighing less than 30 kg. Other pharmaceutical forms may be more appropriate for administration to this population.

Total daily dose of metoclopramide, especially for children and young adults, should not exceed 0.5mg/kg body weight.

Frequency of administration:

A minimum interval of 6 hours between two administrations is to be respected, even if vomiting or rejection of the dose occurs.

Special population

Elderly

In elderly patients a dose reduction should be considered, based on renal and hepatic function and overall frailty

Renal Impairment:

In patient with end stage renal disease (Creatinine clearance $\leq$ 15 ml / min), the daily dose should be reduced by 75%.

In patient with moderate to severe renal impairment (Creatinine clearance 15 – 60 ml / min), the dose should be reduced by 50% (see section Pharmacokinetics)

Hepatic Impairment:

In patient with severe hepatic impairment, the dose should be reduced by 50%.

Contraindication

- Hypersensitivity to the active substance or to any of the excipients listed
- Gastrointestinal haemorrhage, mechanical obstruction, or gastro-intestinal perforation for which the stimulation of gastrointestinal motility constitutes a risk.
- Confirmed or suspected pheochromocytoma, due to the risk of severe hypertension episodes.
- History of neuroleptic or metoclopramide-induced tardive dyskinesia.
- Epilepsy (increased crises frequency and intensity).
- Parkinson's disease.
- Combination with levodopa or dopaminergic agonists
- Known history of methaemoglobinaemia with metoclopramide or of NADH cytochrome-b5 deficiency.
- Use in children less than 1 year of age due to an increased risk of extrapyramidal disorders
- Metoclopramide should not be used during first three to four days following operations such as pyloroplasty or gut anastomosis as vigorous muscular contractions may not help healing

Warning and Precautions

- Avoid doses exceeding 0.5mg/kg/day.
- Extrapyramidal effects, especially dystonic reaction of metoclopramide are more likely to occur in children shortly after initiation of therapy, and usually with doses higher than 0.5 mg per kg of body weight per day.

Special Warning and Precautions for Use

Precautions:

If vomiting persists the patient should be reassessed to exclude the possibility of an underlying disorder e.g. cerebral irritation.

Neurological Disorders

Extrapyramidal disorders may occur, particularly in children and young adults and/or when high doses are used. These reactions occur usually at the beginning of the treatment and can occur after a single administration. Metoclopramide should be discontinued immediately in the event of extrapyramidal symptoms. These effects are generally completely reversible after treatment discontinuation, but may require a symptomatic treatment (benzodiazepines in children and/or anticholinergic anti-Parkinsonian medicinal products in adults).

The time interval of at least 6 hours specified in the section Recommended Dosage should be respected between each metoclopramide administration, even in case of vomiting and rejection of the dose, in order to avoid overdose.

Prolonged treatment with metoclopramide may cause tardive dyskinesia, potentially irreversible, especially in the elderly. Treatment should not exceed 3 months because of the risk of tardive dyskinesia. Treatment must be discontinued if clinical signs of tardive dyskinesia appear. Neuroleptic malignant syndrome has been reported with Metoclopramide in combination with neuroleptics as well as Metoclopramide monotherapy. Metoclopramide should be discontinued immediately in the event of symptoms of neuroleptic malignant syndrome and appropriate treatment should be initiated. Special care should be exercised in patients with underlying neurological conditions and in patients being treated with other centrally-acting drugs. Symptoms of Parkinson's disease may also be exacerbated by metoclopramide.

Methaemoglobinaemia

Methaemoglobinaemia which could be related to NADH cytochrome b5 reductase deficiency has been reported. In such cases, metoclopramide should be immediately and permanently discontinued and appropriate measures initiated (such as treatment with methylene blue).

Cardiac Disorders

There have been reports of serious cardiovascular undesirable effects including cases of circulatory collapse, severe bradycardia, cardiac arrest and QT prolongation following administration of metoclopramide by injection, particularly via the intravenous route.

Special care should be taken when administering metoclopramide, particularly via the intravenous route to the elderly population, to patients with cardiac conduction disturbances (including QT prolongation), patients with uncorrected electrolyte imbalance, bradycardia and those taking other drugs known to prolong QT interval. Intravenous doses should be administered as a slow bolus (at least over 3 minutes) in order to reduce the risk of adverse effects (e.g. hypotension, akathisia).

Renal and Hepatic Impairment

In patients with renal impairment or with severe hepatic impairment, a dose reduction is recommended.

Metoclopramide may cause elevation of serum prolactin levels.

Contains Lactose:

Patients with rare hereditary problems of galactose intolerance, total lactose deficiency or glucose-galactose malabsorption should not take this medicine. Care should be exercised when using Metoclopramide in patients with a history of atopy (including asthma) or porphyria.

Metoclopramide should not be used in the immediate post-operative period (up to 3 – 4 days) following pyloroplasty or gut anastomosis, as vigorous gastrointestinal contraction may adversely affect healing.

Special care should be taken when administering Metoclopramide intravenously to patient with “sick sinus syndrome” or other cardiac conduction disturbances.

There have been very rare reports of abnormalities of cardiac conduction with intravenous metoclopramide. Metoclopramide should be used with care with other drugs affecting cardiac conduction.

Kidney or liver failure

In patients with kidney failure or severe liver failure, a dose reduction is recommended.

Interactions with Other Medicament

Contraindicated combination

Levodopa or dopaminergic agonists and metoclopramide have a mutual antagonism

Combination to be avoided

Alcohol potentiates the sedative effect of metoclopramide.

Combination to be taken into account

Due to the prokinetic effect of metoclopramide, the absorption of certain drugs may be modified.

- **Anticholinergics and morphine derivatives**

Anticholinergics and morphine derivatives may both have a mutual antagonism with metoclopramide on the digestive tract motility.

- **Central nervous system depressants**

(morphine derivatives, anxiolytics, sedative H¹ antihistamines, sedative antidepressants, barbiturates, clonidine and related). Sedative effects of Central

Nervous System depressants and metoclopramide are potentiated.

- **Neuroleptics**

Metoclopramide may have an additive effect with other neuroleptics on the occurrence of extrapyramidal disorders.

- **Serotonergic drugs**

The use of metoclopramide with serotonergic drugs such as SSRIs may increase the risk of serotonin syndrome.

- **Digoxin**

Metoclopramide may decrease digoxin bioavailability. Careful monitoring of digoxin plasma concentration is required.

- **Cyclosporine**

Metoclopramide increases cyclosporine bioavailability (C_{max} by 46% and exposure by 22%). Careful monitoring of cyclosporine plasma concentration is required. The clinical consequence is uncertain.

- **Mivacurium and suxamethonium**

Metoclopramide injection may prolong the duration of neuromuscular block (through inhibition of plasma cholinesterase).

- **Strong CYP2D6 inhibitors**

Metoclopramide exposure levels are increased when co-administered with strong CYP2D6 inhibitors such as fluoxetine and paroxetine. Although the clinical significance is uncertain, patients should be monitored for adverse reactions.

Metoclopramide may reduce plasma concentrations of atovaquone.

Pregnancy and Lactation

Pregnancy

A large amount of data on pregnant women (more than 1000 exposed outcomes) indicates neither malformative toxicity nor foetotoxicity.

Metoclopramide can be used during pregnancy if clinically needed. Due to pharmacological properties (as other neuroleptics), in case of metoclopramide administration at the end of pregnancy, extrapyramidal syndrome in newborn cannot be excluded.

Metoclopramide should be avoided at the end of pregnancy.

If metoclopramide is used, neonatal monitoring should be undertaken.

Breastfeeding

Metoclopramide is excreted in breast milk at low level. Adverse reactions in the breast-fed baby cannot be excluded. Therefore metoclopramide is not recommended during breastfeeding. Discontinuation of metoclopramide in breastfeeding women should be considered.

Undesirable Effects

System Organ Class	Frequency	Adverse Reactions
Blood and lymphatic system disorders		
	Not known	Methaemoglobinaemia, which could be related to NADH cytochrome b5 reductase deficiency, particularly in neonates; Sulfhaemoglobinaemia, mainly with concomitant administration of high doses of sulphur-releasing medicinal products.
Cardiac Disorders		
	Uncommon	Bradycardia, particularly with intravenous formulation
	Not known	Cardiac arrest, occurring shortly after injectable use, and which can be subsequent to bradycardia; Atrioventricular block, Sinus arrest particularly with intravenous formulation; Electrocardiogram QT prolonged; Torsade de Pointes;
Endocrine disorders*		
	Uncommon	Amenorrhoea, Hyperprolactinaemia
	Rare	Galactorrhoea
	Not known	Gynaecomastia

Gastrointestinal disorders		
	Common	Diarrhoea
General disorders and administration site conditions		
	Common	Asthenia
Immune system disorders		
	Uncommon	Hypersensitivity
	Not known	Anaphylactic reaction (including anaphylactic shock particularly with intravenous formulation).
Nervous system disorders		
	Very Common	Somnolence
	Common	Extrapyramidal disorders (particularly in children and young adults and/or when the recommended dose is exceeded, even following administration of a single dose of the drug), Parkinsonism, Akathisia
	Uncommon	Dystonia, Dyskinesia, Depressed level of consciousness
	Rare	Convulsion especially in epileptic patients
	Not known	Tardive dyskinesia which may be persistent, during or after prolonged treatment, particularly in elderly patients, Neuroleptic malignant syndrome
Psychiatric disorders		
	Common	Depression
	Uncommon	Hallucination
	Rare	Confusional state
Vascular disorder		
	Common	Hypotension, particularly with intravenous formulation
	Not known	Shock, syncope after injectable us, Acute hypertension in patients with pheochromocytoma. Transient increase in blood pressure.

* Endocrine disorders during prolonged treatment in relation with hyperprolactinaemia (amenorrhoea, galactorrhoea, gynaecomastia). The following reactions, sometimes associated, occur more frequently when high doses are used:-

* Extrapyramidal symptoms: acute dystonia and dyskinesia, parkinsonian syndrome, akathisia, even following administration of a single dose of the medicinal product, particularly in children and young adults. Drowsiness, decreased level of consciousness, confusion, hallucination

Symptoms and Treatment of Overdose

Symptoms:

Extrapyramidal disorders, drowsiness, decreased levels of consciousness, confusion, hallucination, muscle spasm, itching, urticaria and cardio-respiratory arrest may occur.

Treatment of Overdose

Management

In case of extrapyramidal symptoms related or not to overdose, the treatment is only symptomatic (Benzodiazepines in children and/or anticholinergic anti-parkinsonian medicinal products in adults). A symptomatic treatment and a continuous monitoring of the cardiovascular and respiratory functions should be carried out according to clinical status.

Packing

Tablet: Blister-packs of 10 X 10's / Box
Syrup: 100 ml

Storage Conditions

Store below 30°C in a dry place, protected from direct light.
Emecon Syrup 5 mg/5 ml: Shake well before use.
Keep away from reach of children.

Expiry Date

Tablet: 3 years from date of manufacture
Syrup: 2 years from date of manufacture

Name and Address of Manufacturer:

TERAPUTICS SDN. BHD. (590500-W)
Lot 10 & 11, PERDA, Industrial Park,
Lorong IKS Simpang Ampat B,
14100 Simpang Ampat, S.P.S.,
Pulau Pinang, Malaysia

Name and Address of Product Registration Holder/ Distributor:

ZONTRON PHARMACEUTICALS SDN.BHD. (445695-T)
Lot 10 & 11, PERDA Industrial Park,
Lorong IKS Simpang Ampat B,
14100 Simpang Ampat, S.P.S.,
Pulau Pinang, Malaysia

Malaysian Drug Registration No. :

Emecon Tablet 10 MG: MAL19890263ACZ
Emecon Syrup 5 MG / 5 ML: MAL19890516ACZ

Date of Revision:

16/10/2020